

TEAM ZIBO

STRENGTHENING SEED CONSERVATION INFORMATION AND SUPPORT BETWEEN COMMUNITY SEED BANKS AND FORMAL GENEBANKS IN KENYA

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1. ABSTRACT

Smallholder farmers produce most of Kenya's food and obtain approximately 75–80% of their seed through farmer-managed seed systems (Vernooy et al., 2024). Community Seed Banks (CSBs) play a critical role in conserving these locally adapted seed varieties, while Kenya's national genebank, the Genetic Resources Research Institute (GeRRI), provides long-term conservation through formal documentation, storage, viability monitoring, and safety duplication. Although the 2021 Protocol for Collaboration outlines how Community Seed Banks and the national genebank should work together, there is limited published evidence on how this collaboration operates in practice. Existing studies suggest that interactions are often driven by externally funded projects rather than routine conservation activities (Nyongesa et al., 2026; Vernooy et al., 2024).

This study combines a desk review of Kenya's seed conservation system with field research conducted between 22 and 23 September 2026. Primary evidence was gathered through site visits to the Community Sustainable Agriculture and Healthy Environmental Program (CSHEP) and G-BIACK, together with an interview with the Head of Genebank Information Systems at the Crop Trust. The findings reveal three key challenges. First, essential seed information, including seed identity, source, storage date, and depositor details, is recorded only on handwritten labels attached to seed containers, with no centralized system for long-term record management. Second, seed records do not move with the seed, resulting in repeated germination testing and the recreation of records whenever seed is transferred between organizations. Third, Community Seed Banks have limited visibility into formal conservation support pathways, making it difficult to coordinate conservation activities when seed varieties become vulnerable.

These findings indicate that the primary challenge is not the absence of conservation practices but the lack of a simple, community-controlled system for managing seed information, tracking seed movement, preserving conservation records, and supporting collaboration between Community Seed Banks and conservation institutions. This report therefore proposes **Mbegu Link**, a digital platform that enables Community Seed Banks to maintain a centralized seed registry, manage seed movement, document conservation activities, share knowledge, and coordinate conservation support while preserving community ownership of seed information.

2. INTRODUCTION

Seed is the foundation of every food system. In Kenya, many smallholder farmers rely on locally adapted and open-pollinated seed varieties for food production and agricultural resilience. Community Seed Banks (CSBs) play a vital role in conserving these varieties by collecting, storing, regenerating, exchanging, and documenting seed within their communities. At the national level, the Genetic Resources Research Institute (GeRRI) complements these efforts through long-term conservation, viability monitoring, regeneration, and safety duplication. Together, Community Seed Banks and formal conservation institutions contribute to preserving Kenya's agricultural biodiversity.

Despite their shared conservation goals, these institutions often operate independently, resulting in limited information sharing and coordination. Important information about seed identity, storage history, germination testing, and seed movement is frequently recorded manually or remains fragmented across different organizations. As seed moves between custodians and Community Seed Banks, records are often recreated, testing is repeated, and valuable conservation information can be lost. These gaps make it difficult to monitor seed condition over time, coordinate conservation activities, and respond promptly when seed varieties become vulnerable.

This study investigates how Community Seed Banks in Kenya manage seed information, monitor seed condition, facilitate seed movement, and respond to conservation challenges. Through a combination of literature review and field research, the study identifies the information and coordination gaps that affect effective seed conservation and explores how digital technology can support existing conservation practices without replacing them.

Based on these findings, this report proposes **Mbegu Link**, a community-controlled digital platform designed to improve seed information management and conservation coordination. The proposed minimum viable product (MVP) focuses on four core functions: **Seed Registry**, **Seed Movement**, **Knowledge Sharing**, and **Conservation Attention & Coordination**. By strengthening record management, improving the traceability of seed movement, facilitating knowledge exchange, and supporting timely conservation decisions, Mbegu Link aims to enhance collaboration among Community Seed Banks while preserving community ownership of seed information.

2.1 Research Goals

- Understand how Community Seed Banks conserve, store and manage seed varieties.
- Understand how Community Seed Banks identify poor storage, seed deterioration and declining viability.
- Find out what information Community Seed Banks record (identity, quantity, storage, viability, regeneration).
- Understand what happens when a conservation problem is identified and how support is accessed.
- Under how Community Seed Banks manage and store seed testing information.
- Understand how safety duplication is handled and whether backup information is recorded.
- Identify what information and support Community Seed Banks need before deciding whether a digital solution is appropriate.

2.2 Research Questions

Main research question

How do community seed banks in Kenya identify, manage and respond to seed conservation risks, and where are the information and coordination gaps between community seed banks and formal conservation institutions?

Supporting questions:

1. How do Community Seed Banks currently store, manage and monitor their seeds?
2. How do they know when a variety needs attention, testing or regeneration?
3. What do they record each seed variety, and how often is it updated?
4. What happens when a problem is identified, and who is contacted?
5. Is there an established route to support, or does it depend on projects and personal relationships?
6. How is safety duplication understood and recorded?
7. What information should remain under community ownership and control?

2.3 Research Approach and Evidence Base

The study employed both qualitative and quantitative research methods, including semi-structured interviews, observation of seed storage practices, review of available records where permitted, and an online survey. Field research was conducted between **22 and 23 September 2026**. The evidence presented in this report is drawn from:

- A structured desk review of Kenyan and international literature.
- Recorded and transcribed observations from site visits to the **Community Sustainable Agriculture and Healthy Environmental Program (CSHEP)** and **G-BIACK**, two community seed conservation organizations.
- A recorded interview with the Head of Genebank Information Systems at the Crop Trust, who oversees the **Genesys** and **Genebank Global Community Edition (GGCE)** systems.

3. PROBLEM STATEMENT

While community seed custodians utilize practical local methods to manage seeds, vital information remains informal and highly dependent on individual memory or fragmented project relationships. Consequently, when seeds move between diverse stakeholders or institutions, this discontinuity makes a variety's history and genetic identity difficult to reconstruct. This missing information layer severely complicates critical decisions regarding testing, seed regeneration, duplication, and formal conservation support from central entities like GeRRI, ultimately leaving vulnerable crop varieties at risk of unnoticed decline or extinction

4. LITERATURE REVIEW

4.1 Existing Research

Seed systems and food security in Kenya

Agriculture accounts for about one fifth of Kenya's GDP and employs more than 40% of the population, dominated by smallholders. Muthoni and Nyamongo (2008) show that formal seed dominates commercial crops in high-potential zones, while orphan crops such as millets, sorghum, and cowpeas rely overwhelmingly on farm saved seed. Seed exchange in these systems rests on trust and social relationships.

Community seed banks

Nyongesa et al. (2026) studied 18 Community Seed Banks from a pool of 107 across 11 counties. Membership grew 148% between 2018 and 2024, women made up over 80% of members, and more than 80% of CSBs had moved into value addition. The same study documents weak record-keeping and suboptimal seed handling, and warns that its sample limits generalization. Vernooy et al. (2024) identify five

sustainability strategies, including partnerships with national genebanks and low-cost seed quality technologies.

Methodological Synthesis Note: While a comprehensive evaluation of current academic and institutional literature confirms that no reviewed source demonstrates an operational system combining identity, age, quantity, viability, and duplication status for community collections, this finding is treated strictly as a gap in documented baseline literature. It does not definitively indicate that no such system exists anywhere globally, but rather underlines a critical lack of open-source frameworks or accessible scholarly models bridging informal and formal conservation data.

Seed storage and viability

Whitehouse, Hay, and Lusty (2020) and Zhang et al. (2021) explain that moisture and temperature drive seed aging, and that germination testing is the standard viability method. Wijnker et al. (2025) show that even professional testing carries uncertainty. With a 50-seed sample and a true germination of 90%, there is a 12.2% chance that the result falls below an 85% threshold by sampling variation alone. FAO Genebank Standards set formal thresholds (for example, regeneration when viability falls below 85% of initial value), which were written for formal genebanks and should not be imposed on Community Seed Banks unchanged.

Formal genebank conservation

Genebanks follow a structured workflow from acquisition through documentation, storage, viability monitoring, and regeneration to safety duplication. GeRRI, operating under KALRO, has supplied data to the global Genesys Platform for several years and has backed up material at the Svalbard Global Seed Vault.

Law and policy

Kenya's Seeds and Plant Varieties Act regulates commercial seed activities and was challenged by 15 smallholder farmers. On November 27, 2025, the High Court at Machakos declared several challenged provisions unconstitutional (*Wathome & 14 others v KEPHIS & another*). The government has appealed, and the final judicial or legislative outcome remains unestablished in the material reviewed.

Data Standards, Biocultural Rights, and Governance Frameworks

For the proposed digital architecture (Mbegu Link), the data model is fundamentally not a secondary technical or software concern; the data model is explicitly designed

as an active part of the conservation intervention itself. Bridging community workflows with formal genebanks requires deep integration with established data standards and rights frameworks, structured across four dimensions:

- **Accession, Passport Descriptors, and Taxonomy:** Rigorous biological conservation requires strict alignment with taxonomic classifications alongside standardized tracking syntax. Mbegu Link integrates the international Multi Crop Passport Descriptors (MCPD) standards developed by Bioversity International and FAO. This standard provides the exact infrastructure needed to map a seed sample's biological pedigree from original community acquisition through long-term preservation.
- **Persistent Identifiers, DOIs, and Provenance:** To prevent the systemic information fragmentation that occurs when seeds move between diverse stakeholders, modern data registries rely on Digital Object Identifiers (DOIs) or permanent unique keys. Securing a strict digital chain of provenance recording geographic coordinates, collection dates, and historical custody ensures that an accession's identity remains completely intact during cross branch transfers or localized re-testing.
- **Traditional Knowledge and Indigenous/Local Naming:** Informally managed crop varieties carry deep cultural history. Data architectures must preserve local and indigenous variety naming conventions as first-class metadata attributes alongside scientific taxonomy. This ensures that community-led classifications and traditional ecological knowledge are structurally preserved, valued, and digitized without being overwritten by formal institutional vocabularies.
- **Access and Benefit Sharing, Community Consent, and Data Sovereignty:** Digitizing community collections directly intersects with indigenous data sovereignty and intellectual property protections. In alignment with international frameworks like the Nagoya Protocol on Access and Benefit-Sharing, Mbegu Link treats community consent as a core structural dependency. The underlying data model must protect community rights, ensuring that information visibility facilitates collaborative conservation support from central national entities like GeRRI without leading to unauthorized resource extraction or exploitation.

4.2 Case Insights

The literature and our field evidence show a wide range of connections between Community Seed Banks and formal institutions.

- **Connected end:** GeRRI distributed 51 sorghum varieties to 511 farmers in Busia and Siaya, working with Rongo University and the Seed Savers Network.

The Alliance of Bioversity International and CIAT and GeRRI established four Community Seed Banks in Kisumu, Vihiga and Kericho, and the Pap Kadundo Integrated Community Seed Bank was launched in Kisumu in March 2026. In each case a funded project built the link deliberately.

- **Disconnected end:** the Ambira Community Seed Bank in Siaya holds dozens of varieties and has no working relationship with GeRRI.
- **Our field site:** the sites visited work mainly with farmers, under an umbrella body PELUM Kenya, and with sister organisations. The sites visited have not deposited seeds at the genebank.

The 2021 Protocol for Collaboration is the formal bridge, but the version we reviewed has blank signature fields and no systematic account of its implementation.

4.3 Existing Solutions

System	What it does	What it does not document
Genesys	Global public portal aggregating genebank passport data (identity, origin, biological status, storage type, safety duplication)	Seed viability is not shared publicly (Crop Trust interview) not built for community level conditions.
GGCE / GRIN-Global	Internal genebank management: inventory, viability, storage location, requests	Used inside the genebank by staff and not a shared community system
Seed Savers Network platform and Seed Exchange Application	Networking, seed exchange, training resources	Public descriptions do not show storage, viability or duplication tracking
Seed Exchange Kenya	Transactional seed access	No conservation monitoring described
Mbeu Yedu (Zimbabwe)	CSB discovery, borrowing, registry	No conservation risk monitoring described

System	What it does	What it does not document
Physical tools in Community Seeds Banks	Hygrometers, zeolite beads, airtight containers, solar dryers	Do not produce or store any records of the seeds.

4.4 Research Gap

1. No shared record of the stored seed condition.
2. No reviewed source shows a system combining seed identity, age, quantity, viability and safety duplication status for community collections.
3. Unknown implementation of the collaboration protocol. What information flows between Community Seed Banks and GeRRI, and what happens when a collection becomes vulnerable, is undocumented.
4. Sample identity across institutions. We found no literature on whether identifiers used by a Community Seed Banks and by a genebank can be linked to the same seed material. The Crop Trust interview identified this as the issue researchers should examine.
5. Community appropriate seed testing. No validated low-cost testing protocol exists for Community Seed Banks, and formal thresholds cannot simply be transferred.
6. Unproven need for usage of technology e.g sensors and basic AI monitoring tools. It is unclear whether continuous data recording from monitoring technology adds value to the Community Seed Banks operations.

5. CASE STUDY ANALYSIS

Description of the Selected Case Study

The case study covers Kenyan community seed conservation operations at the field research sites visited and across the network around it. The first site visited was the Community Sustainable and Agriculture Environment Program (CSHEP) , an organic farming and seed conservation organisation under the PELUM Kenya network. Their site is located in Kiserian southwest of Nairobi in Kajiado County, they are currently storing beans, maize, amaranth, flax, cumin, soya beans and several tree and herb seeds, alongside value-added products such as composite flours and solar drying of fruits. The second site visited is G-BIACK in Thika, Kiambu County, it is a community youth sustainable training center and seed conservation organisation, it works with Seed Savers Network Kenya and is under the PELUM

network. The case is set against the national genebank (GeRRI), which holds over 51,800 accessions and shares passport data through Genesys.

Comparative Analysis

Dimension	Site visited	Formal genebank (GeRRI / Genesys)
Identity record	Handwritten labels which have the following information; seed type, source, storage date, contact.	Accession number (e.g. GBK-000001) with passport data and DOI
Names	Local names (e.g. a Kikuyu name), names adopted from the source organisation. English or scientific names are sometimes unknown.	Taxonomic names and standardised descriptors
Database	None present, the interviewed participants said there was no record book except the label stuck to the containers holding the seeds.	GGCE for internal managementGenesys for public data
Storage	Room at 21–25°C, glass bottles and jars. All seeds are stored in one place	Controlled temperature and humidity, base collections around −18°C
Viability testing	Petri dish and cotton wool or tissue germination test, about five seeds per dish and are stored in the same room but at different positions. The seed growth is tested on a scale of 75% to 100% using manual observation.	Standard germination tests with defined sample sizes and thresholds
Regeneration trigger	Plant the oldest stock first. The stored seed storage is expected to last about three years or more.	Regenerate when viability drops below a set threshold
Safety duplication	Multiply endangered seed is distributed to farmers to grow at home	Formal duplicate at a second site (e.g. Svalbard)

Sharing	Physical seed sent with a note and recipients have to re-test and re-record the seeds received	Data submitted to Genesys under a formal letter of agreement

Stakeholders Involved

- **Seed custodians, coordinators and committee members:** hold, test and label the seeds.
- **Seed networks:** PELUM Kenya (organic farming umbrella body) and Seed Savers Network Kenya provide training and coordination.
- **National genebank:** GeRRI (KALRO) is the formal conservation and documentation authority. Dr Nyamongo has been contacted for an interview.
- **Global information partners:** the Crop Trust supports Genesys and GGCE and has a long standing data partnership with GeRRI.
- **Regulators, funders and researchers:** KEPHIS, county departments, Alliance Bioversity–CIAT, FAO, GIZ and universities.

Pain Points Observed

- Identity is fragile, seed arrives with local names, and the English name is usually unknown. A label on a bottle is the only record.
- Nothing is recorded outside the bottle. Test results, sources and dates are not kept anywhere else.
- Re-testing at every transfer. Each recipient tests the seed again because there is no record on how long the seed has been stored or what the last results were.
- No route to the genebank. The participant did not know how to deposit seed or what quantity is required.
- Support follows relationships. Help and training arrive through PELUM Kenya, sister organisations and Seed Savers Network.
- Backup is informal and invisible. Redistributing endangered seed to farmers is a real safety measure but is not recorded as one.

Existing Solution Limitations

Formal systems such as Genesys and GGCE are strong at what they cover, but they sit inside institutions and capture identity and formal storage data rather than community-level conditions. Community platforms concentrate on exchange and networking. The 2021 Protocol describes the relationship but does not supply a tool for the routine exchange of information. Physical technologies such as hygrometers help custodians but leave no record that a network or genebank can see.

6. FINDINGS / KEY PAIN POINT IDENTIFICATION

These initial findings come from a localized contextual inquiry involving two site visits, one expert interview, and public GeRRI records. To ensure methodological rigor and protect against over generalizing, these insights are explicitly presented as baseline evidence to be tested and validated through a wider comparative sample, rather than final conclusions about all Community Seed Banks (CSBs) in Kenya.

6.1 Methodological Context & Required Additional Sampling

The fieldwork successfully captures granular, real world workflows regarding seed containers, local labels, storage conditions, and transfer practices. However, this small sample size remains insufficient to establish the broader prevalence of these workflows across the sector. Crucially, the target sites may differ significantly from CSBs that maintain stronger, more structured institutional relationships.

To transform these initial insights into a robust, quasi-comparative research design

capable of isolating how institutional connectivity alters CSB operations, the next phase of this study will deploy an expanded sampling framework consisting of:

- **3–5 CSBs** characterized by weak or completely absent GeRRI relationships.
- **3–5 CSBs** maintain active, formal institutional relationships.
- **2–3 distinct seed networks** or regional intermediaries.
- **2–4 GeRRI / national genebank** staff members.
- **3–5 individual farmers and seed custodians** selected per participating CSB.

6.2 Seed Information Exists, but Only on the Container

The visited site labels each bottle with four specific parameters: seed type, source, date of storage, and the depositor's contact information. When a team member asked whether a centralized system existed for entering and preserving this data, the participant replied that they were **"just doing it manually"** and did not maintain **"any record"** beyond the physical bottles. This isolated case confirms the localized record-keeping weaknesses previously described by Nyongesa et al. (2026) and aligns with proposition **P1** (that seed condition and viability information is frequently incomplete or inconsistent).

6.3 Meaningful Local Practices Lack a Recording Foundation

The field team observed several highly meaningful local workflows that should be built upon rather than replaced. These include:

- **Inventory Rotation:** New deposits are kept in separate containers by packaging date, ensuring the oldest seed is planted first.
- **Visual Quality Control:** Clear glass and plastic jars allow staff to quickly inspect, identify, and discard visibly damaged seeds.
- **Organic Preservation:** Natural protectants (such as diatomaceous earth, *Lantana camara*, and wood ash) are used for seed preservation. Seed intended for consumption is thoroughly washed before milling.
- **Actionable Germination Rules:** A strict viability protocol is followed: batches achieving above 75% growth are approved for long-term storage, while batches falling below 75% growth are diverted to immediate consumption.

These observations directly support proposition **P2**: the core operational vulnerability is not an absence of community monitoring, but rather that what is successfully observed on-site is never formally recorded, digitized, or escalated.

6.4 Every Inter-Site Transfer Restarts the Record

When seed is transferred to another community branch such as G-BIACK in Kiambu it travels accompanied only by a basic paper note. Upon arrival, the receiving branch must run its own independent germination tests and log the details from scratch. The participant explained this duplication is mandatory because they have no way of knowing how long or under what conditions the seed was stored at the prior location.

The immediate consequence is duplicated testing, wasted seed stock, and a completely fragmented history. This serves as a micro-level version of the systemic risk raised during our expert interview: genetic identity and historical integrity break down the moment material moves between separate players.

6.5 Formal Genebank Contact is Locally Absent

When asked, *"How is seed formally deposited at a national genebank?"* the participant stated they had never done it, did not know the required administrative process, and guessed that "a few kgs" might be required. The site's operational focus remains strictly confined to farmer-to-farmer and organization-to-organization exchanges.

While this aligns with proposition **P4** (that formal genebank interaction is triggered sporadically by isolated projects and relationships rather than a routine, institutionalized pathway), it must be rigorously tested across the wider sample before generalizing, given that entities like Katambasye and G-BIACK have documented ties to formal institutions.

6.6 Safety Duplication is Informal and Invisible

To preserve crop varieties at high risk of localized extinction, the site multiplies the seed, distributes it to trusted area farmers to grow and store, and encourages select individuals to independently take samples to the national genebank. While this functions as an admirable form of decentralized, distributed backup, it remains completely undocumented. No centralized, verifiable record exists to show which specific varieties are duplicated or where they are located, supporting proposition **P5** regarding the critical lack of visibility surrounding duplication status.

6.7 Storage Infrastructure is Bound to Ambient Conditions

Seed stocks are kept in a standard, uncooled room at ambient temperatures averaging 21–25°C, with the participant noting they have no access to cold-storage technology. Because formal genebank preservation standards are drastically stricter, the site is forced to limit its storage horizons to roughly three years, anticipating a sharp decline in germination viability after the second year.

6.8 Formal Genebank Data Structure vs. The Identity Link

An interview with a Crop Trust expert clarified that while formal genebanks maintain highly structured internal data (seed count, 100-seed weight, viability, precise room location) within GGCE, they only publish a limited subset to external discovery platforms like Genesys (specifically: passport identity, storage type, safety duplication status, and images). Viability percentages are intentionally withheld from public views. The expert highlighted three key areas of relevance to this study:

- **The "Data Archaeology" Challenge:** The most vital historical data often remains locked on physical paper. Historical collecting mission forms containing original farmer names and coordinates exist but are incredibly costly and time-consuming to digitize.
- **The Spreadsheet Gap:** The expert noted that common data gaps are frequently caused by reliance on manual Excel spreadsheets rather than specialized genebank management systems, creating a significant "time gap" between initial data acquisition and public availability. *Note: This is presented*

strictly as an expert professional opinion rather than an established universal finding.

- **The Identifier Disconnect:** The primary operational question is passport linkage. Researchers must verify whether GeRRI records capture the specific local identifiers used by CSBs, and conversely, whether CSBs record and preserve the unique accession numbers assigned by GeRRI.

A preliminary inspection of GeRRI's public records on Genesys aligns with this concern. The reviewed accessions (**GBK-000001 to GBK-000106**, representing sorghum collected in 1987 and 1996) carry clear taxonomy, coordinates, and collection dates. However, critical metadata gaps exist: several records lack elevation metrics, at least one record (**GBK-000011**) completely lacks a georeference, and only 251 out of approximately 51,832 total Kenyan accessions include images. Most critically, the public table view fails to display any community seed bank cross-identifiers. Because we cannot confirm from this view alone that no linkage exists, this must be cross-checked against a full Multi-Crop Passport Descriptor (MCPD) data download and verified directly with GeRRI staff.

6.9 Perceptions Requiring Field Verification

During interviews, the participant asserted that Kenyan law legally requires an active Community Seed Bank connection before any seed can be sold. This belief contradicts current seed policy literature, which restricts formal certification and registration mandates strictly to commercial-scale seed enterprises rather than informal CSB networks. The team has recorded this as a localized participant belief to be formally verified. More broadly, the divergence between expected desk research assumptions and these raw field realities underscores exactly why this field validation stage is mandatory.

6.10 Summary of Key Pain Points

	Pain point	Evidence	Related proposition
1	Seed identity and history held only on bottle labels	Site visit	P1
2	Practices exist but are not recorded or escalated	Site visit	P2

3	Re-testing and re-recording at each transfer	Site visit	P1, P6
4	No known route to the national genebank	Site visit and literature	P4
5	Duplication is informal and invisible	Site visit and Crop Trust	P5
6	Identifier linkage between CSB and genebank unverified	Crop Trust and Genesys	P3, P4
7	Paper and spreadsheet records slow data availability	Crop Trust (opinion)	P6

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7. PROPOSED SOLUTION

Mbegu Link: A Community-Controlled Information Link for Seed Conservation

Mbegu Link is a platform that gives Community Seed Banks (CSBs), seed custodians and the national genebank (GeRRI) one community-controlled seed record that stays connected as seed moves. It makes seed identity, condition, movement and support needs visible, so that communities can conserve seed over the long term and reach the right institution when they need help. It strengthens the link between CSBs, farmers and formal conservation institutions such as the Genetic Resources Research Institute (GeRRI).

The research found that important information about seed identity, viability, storage and movement is recorded manually, often on bottle labels, and is hard to trace when seed changes hands. Mbegu Link provides a structured record that the community controls, and a clear pathway for requesting storage, duplication, support or formal conservation.

A Knowledge Hub also lets CSBs record and share their practical knowledge, and choose whether each entry stays private, is shared within their institution, or is made public.

The platform complements, and does not replace, genebank systems such as GGCE and Genesys. It is the information bridge between community conservation and formal institutions. It is a decision-support tool: it records, reminds and flags, while people make the conservation decisions.

MVP scope. The MVP covers four areas: seed registry, seed movement, knowledge sharing, and conservation attention and coordination. It serves three types of user: seed custodians (mobile app), CSB coordinators (PWA) and GeRRI seed conservation specialists (PWA). Where a feature is planned for after the pilot, the section says so.

7.1 Seed Registry

A custodian registers a seed variety on the mobile app, and the CSB coordinator manages the master registry for the whole collection. Each seed record is built from what CSBs already capture:

- Local and community name of the seed variety
- Crop name, and variety and scientific name where known
- Source: the name and phone number of the person or organisation that supplied or deposited the seed
- Quantity
- The CSB responsible for the record, and the custodian currently holding the seed
- Germination and viability history, kept as dated events

Lot changes. Seed is biological material, not ordinary stock. Regeneration and safety duplication are recorded as dated events on the seed record, so earlier results are never overwritten.

After the pilot. A Mbegu Link ID linked to the community's own identifier and the genebank accession number (Community Identifier → Mbegu Link ID → Genebank Accession Number), and lot lineage with a parent lot, a generation and the event that created it, so that split, merged and regenerated lots are traceable.

7.2 Record Trust and Provenance

Six fields are treated as first-class parts of a record and are shown wherever the record is shown. Some are already provided by the registry and events, and the rest are added after the pilot.

Field	Meaning	Status
Last observed	When someone last physically saw the seed	After the pilot
Last tested	Date of the most recent germination or viability test	Taken from the latest test event
Last updated	When the record was last edited	After the pilot
Data owner	The CSB responsible for the record's accuracy	The owning CSB
Current custodian	Who holds the seed now	The recorded holder
Confidence / provenance	How the information was obtained and how certain it is	Source is recorded; confidence rating after the pilot

Records display their age clearly (for example, "last tested 14 months ago") and are marked as possibly out of date once past a threshold agreed with the community. Nothing is deleted when it goes stale; it is flagged for review.

7.3 Conservation Monitoring

CSB coordinators track the testing history of every seed in their collection and record germination test results: date, number of seeds tested and number that germinated. Regeneration and safety duplication are recorded in the same history. Each entry is a new row, so a falling germination result stays visible.

Coordinators also configure the baseline parameters for their CSB: the germination threshold (a community-agreed value, not an imported genebank standard) and the maximum seed-holding period. These drive the attention flags in 7.4.

A corrected test is entered as a new event, and the original entry is preserved in the history. Reminders and alerts support human judgement; they do not replace custodians or laboratory personnel.

After the pilot. Scheduled testing reminders, storage observations, and marking corrected entries as superseded.

7.4 Attention Dashboard (Coordinator View)

The CSB coordinator can see the seed varieties or lots that need attention, based on the configured thresholds and the recorded conservation information. For example:

Germination below configured threshold → Seed requires attention → Coordinator reviews record → Coordinator decides appropriate action

Possible actions include:

- Conduct another germination test
- Regenerate the seed
- Share or transfer the seed to another CSB
- Request GeRRI for seed or lot conservation assistance
- Coordinate safety duplication
- Continue monitoring the seed variety or the lot

Flags: germination below the configured threshold, seed held longer than the maximum holding period, and requests waiting for a decision.

Flags after the pilot: overdue viability testing, missing or incomplete records, records not observed for a long time, varieties held at only one location, missing safety duplication, and transfers not yet confirmed.

Each flag is a prompt for human review with the underlying records one tap away. Flags never state a conclusion about seed health.

7.5 Seed Movement: Loans, Sharing and Transfers

Custodians submit structured requests to borrow seed varieties. Coordinators review and manage requests for seed sharing, loans, transfers and movement coming from custodians and other CSBs. Each request moves through defined stages:

Pending → Approved or Declined → In Progress → Completed (received)

Loans have a return date, and a loan is marked Returned when the seed comes back. The sender stays the recorded custodian until the receiver confirms receipt. The workflow also handles cases that happen in practice:

- **Not confirmed within an agreed period::** the request stays In Progress and appears on the coordinator's list of outstanding requests.
- **Lost in transit::** custody stays with the sender, and the coordinator notes the exception on the request.
- **Partial transfer::** the quantity received is recorded alongside the quantity sent, and the remainder stays with the sender.
- **Declined::** the receiving CSB can decline with a stated reason.

Each request records the requester, the receiving CSB or GeRRI, the seed record, the type (loan, transfer or support), quantity, details, due date, status and time of last change.

7.6 Conservation Attention and Support Request Pathway

When a seed variety is critically threatened, the coordinator flags it and submits an assistance request to GeRRI specialists, or coordinates with another CSB for safety duplication and longer-term conservation. A CSB can ask for help with: additional storage, safety duplication, declining viability, identification or documentation, transfer to another facility, or technical support.

GeRRI specialists review each request and can approve, reject or coordinate it. The request is tracked from Pending through Approved or Declined, In Progress and Completed. An institution can decline or ask for more information, and the CSB sees the reason. A specialist sees only the seed records that the CSB has linked to a support request. If GeRRI rejects a submission, the lot and its record stay with the CSB, unchanged, and the outcome is logged. This replaces reliance on individual contacts with a repeatable pathway.

7.7 Community-Controlled Sharing, Privacy and Security

Seed information can include farmer identities, indigenous knowledge, collection locations, provenance and potentially valuable genetic resources. Not all of it should be discoverable. Community control is an architectural requirement, built into the data model and enforced on the server, not a setting in the interface.

- Role-based access enforced server-side, so hiding something in the interface is never the only protection. Each user belongs to one organisation, which keeps each CSB's data separate.
- Verified accounts: custodians join with a CSB invitation code and a phone OTP. Coordinators and specialists are registered by the Mbegu Link team, and an organisation must be verified before its users can be activated.
- GeRRI specialists see only the seed records a CSB has linked to a support request.
- A CSB's location is shown to other users only if the CSB has agreed to share it.
- Knowledge entries are private by default, and making one public needs deliberate confirmation.
- Audit logs of security-relevant actions, such as sign-ins, failed sign-ins, invitation code use, permission denials, sharing changes, exports and seed transfers. Logs are append-only and never contain passwords or tokens.
- Passwords stored only as hashes, and encryption of data in storage and in transit.
- Data minimisation, permission-checked and logged exports limited to shared data, and backup and recovery with tested restore procedures.

After the pilot:

- Consent records: what was shared, with whom, for what purpose, when and who agreed.
- Revocable sharing that takes effect as soon as it is made, with copies already held by an institution handled according to the agreement recorded at the time of sharing.
- A fourth sharing level for seed networks, and masking of exact coordinates as a coarse area (for example, sub-county) once coordinates are collected.
- A deletion and retention policy agreed with communities, including what happens to records if a CSB leaves the platform. This should be agreed before any data is shared.

7.8 Knowledge Hub

Alongside seed records, CSBs hold knowledge: how a variety is grown, stored and prepared, its uses, seasonal practice, seed-saving techniques and lessons learned. Much of this is held by individual custodians and elders and is lost when people move on. Custodians can read this knowledge in the app to learn more about seed varieties, and CSBs can record it and choose who can see it.

What can be shared: short written entries with photos or videos, in local languages as well as English or Kiswahili. Files are kept in Media Storage and the database holds only their link. Entries can be linked to a variety and tagged by crop, practice or topic.

Visibility is chosen for every entry, and the default is private:

- **Private (default)::** visible only to the author and their CSB.

- **Organisation::** visible to members of the organisation that owns the entry.
- **Public::** discoverable and readable by anyone.

A network level, visible to participating CSBs and coordinators, will be added with the Network Coordination Portal. Safeguards, because knowledge can be sensitive:

- Making an entry public is a deliberate step with a warning, a named approver in the CSB, and a consent record. Nothing becomes public by default or by accident.
- The warning states that once knowledge is public, copies can be made that the CSB cannot recall. Withdrawing an entry removes it from the hub but cannot undo copying.
- Entries that name farmers, describe community-specific or ritual knowledge, or reveal exact collection locations are flagged before publishing.
- Each entry records who contributed it, and the contributor can choose to be credited or anonymous.
- Access follows the same role-based rules, audit logging and revocation as seed records (7.7). The CSB can edit, restrict or withdraw its own entries at any time.
- Public knowledge sits in a separate space from seed record data, so publishing a guide never exposes the underlying seed record.

Freshness and sustainability. Each entry shows who wrote it, when it was created and when it was last reviewed, and entries not reviewed for a long time are marked as possibly out of date, as with seed records (7.2). The hub only helps if people use it, so entry forms are short, entries can be photo-first. The pilot should test whether CSBs actually contribute and read before the hub is scaled.

7.9 Role-Based Platform Structure

A shared data model serves three interfaces.

Seed Custodian (Mobile App). Create and register an account with contact details and a local community or regional affiliation. Register a seed variety for storage with its local name, sample quantity and CSB location. Submit structured requests to borrow seed varieties. Access knowledge about seed varieties.

Community Seed Bank Coordinator (PWA). Verify the institution profile and set up the CSB portal to manage its local inventory. Manage the master registry: track the physical seed holdings and testing histories, and record germination test results. Configure the germination threshold and maximum seed-holding period. Monitor seed movement and review requests for sharing, loans, transfers and movement. Initiate conservation support requests for threatened seed.

GeRRI Seed Conservation Specialist (PWA). Log in to the secure institutional dashboard with verified GeRRI credentials and assigned roles. Review seed and conservation information shared by participating CSBs. Review, approve, reject or coordinate assistance requests.

Coordinate safety duplication, transfer or longer-term conservation when a CSB needs institutional support.

After the pilot. A Network Coordination Portal for seed-network coordinators to view information shared by participating CSBs, monitor support requests, coordinate between CSBs and institutions, and moderate network knowledge spaces. GeRRI specialists can also assign or record accession numbers and link them to the community identifier.

7.10 Four Connected Modules

The MVP is organised into four connected modules, one for each area of scope.

Module	Purpose
1. Seed Registry	Seed records with local and alternative names, source, quantity, the responsible CSB and the current holder, plus the testing history of each seed (germination tests, regeneration and safety duplication)
2. Seed Movement	Loan, sharing and transfer requests, request review, and custody through each movement
3. Knowledge Sharing	Knowledge Hub entries with private, organisation and public visibility, photos and videos, and withdrawal
4. Conservation Attention & Coordination	Threshold-based attention flags in the Coordinator view, support requests to GeRRI and CSBs, and GeRRI review and response

7.11 Technical Approach

The technically difficult parts are not the registry itself. The hard problems are identity, sustained data entry and institutional acceptance, and technical research should focus on them.

Frontend: a mobile app for custodians, and a Progressive Web App for coordinators and GeRRI specialists, both used with an internet connection. The choice of an app over paper or a spreadsheet is tested in the pilot (Section 9) and not assumed.

Concurrent edits: two users can edit the same record, or submit the same request twice. Conflicts are shown to a person and never silently overwritten, and history is append-only.

Backend: a FastAPI service over a relational database with eight tables (users, organisations, seed registry, events, requests, knowledge entries, media files and audit logs). Photos and videos are kept in Media Storage, and the database holds only their links. The backend writes the audit log for every module.

Access control and security: as set out in 7.7, enforced on the server.

Integration: an export and API layer for future exchange with GGCE and Genesys, subject to confirming the technical pathway with the institutions concerned.

7.12 Expected Value

- Seed traceability from community record to formal conservation record
- Less repeated testing, because history travels with the seed
- Better monitoring through structured records and threshold-based attention flags
- Visibility of risks such as seed below the germination threshold, seed held too long and missing backups
- Clear custody during loans and transfers, including when things go wrong
- A defined support pathway between CSBs and GeRRI
- Community-controlled information sharing built into the system
- Community knowledge kept and shared on the community's terms, instead of lost when custodians move on
- A stronger link between community and formal conservation without replacing either

7.13 Proposed MVP

Goal. CSBs, seed custodians and GeRRI keep one community-controlled seed record that stays connected as seed moves. Seed identity, condition, movement and support needs become visible, so communities can conserve seed over the long term and reach the right institution when they need help.

Scope. 1. Seed registry. 2. Seed movement. 3. Knowledge sharing. 4. Conservation attention and coordination.

Interfaces. Seed Custodian mobile app; CSB Coordinator PWA; GeRRI Seed Conservation Specialist PWA (see 7.9 for what each user can do).

Workflows. Account registration with community or regional affiliation; seed registration with local names; germination test recording with corrections; threshold and holding-period settings; loan, sharing and transfer requests with custody confirmation; conservation support requests and GeRRI responses; attention flags in the Coordinator view; knowledge entries with three visibility levels, consent record and withdrawal; role-based access and audit logs.

Public knowledge. Public visibility can be switched off for the pilot until consent rules are agreed with communities.

MVP success.

- A local seed custodian can securely register and update a threatened seed variety on the mobile app.
- CSB coordinators can track seed identity, condition, testing history, custody, movement and conservation attention, while managing seed-sharing and assistance requests.

- When a seed variety needs institutional support, the CSB can flag the conservation concern and connect with GeRRI specialists to coordinate safety duplication or longer-term conservation before the seed becomes extinct.

7.14 Critical Test Cases

Because biological inventory is not ordinary inventory, the following cases must be tested before any pilot:

Area	Test cases
Identity	Duplicate seed record; two local names for the same material; same name for different material; unknown variety or scientific name; missing source; conflicting IDs
Lot changes	Seed regenerated and recorded as an event; seed shared to another CSB; later: split and merged lots, new generations
Thresholds	Germination result exactly at the threshold; threshold changed after tests were recorded; seed passing the maximum holding period; seed with no test recorded
Movement	Request never confirmed; seed lost in transit; partial transfer; request declined with a reason; loan not returned by the due date
Data quality	Incorrect storage date; germination test entered incorrectly; wrong sample size; historical test corrected; deleted record
Concurrent use	Two users editing the same record; the same request submitted twice; connection lost while saving
Access and institutions	Invitation code used for the wrong CSB; unverified organisation activating users; specialist trying to open a record not linked to a request; GeRRI rejects a request; accession number changed or corrected
Knowledge Hub	Entry published as public by mistake; entry withdrawn after being public; entry naming a farmer or exact location; visibility changed from public back to private; contributor asks to be anonymous; photo or video upload fails or is interrupted

8. DISCUSSION OF PROPOSED SOLUTION

Evidence limits. The findings rest on two site visits, one expert interview and public records. The sites visited are connected to a network but not to a genebank, so they may not represent CSBs with stronger institutional ties.

A deliberately small MVP. The MVP covers four areas and three users. Features that depend on agreements not yet made, such as consent records, revocable sharing, the network portal and accession-number management, are held back until the pilot shows what is needed.

Testing, not assuming, the need for technology. We deliberately avoided a technology-first approach. Moving records onto a platform does not by itself improve conservation outcomes. A pilot comparing paper, spreadsheet and app will show whether structured records change behaviour before software is built at scale.

Sustainability is the main risk. Records only help if somebody maintains them, and volunteers already carry heavy workloads. A database that stops being updated after three months is worse than a bottle label, because users may assume it is current. The response is design-based: short forms, assigned responsibility, and visible record age and provenance (7.2), so that stale information is recognisable as stale.

Hard technical problems. Identity across institutions, sustained data entry and institutional acceptance are the real difficulties, not building the registry. These, and not the interface, should dominate technical research.

Uncertainty in viability data. Germination results from small community samples are noisy (Wijnker et al., 2025). Any flag prompts human review and does not deliver a verdict. The 75% rule used at the site is a local practice and not a validated standard. In the platform it is only a starting value, and each coordinator sets the threshold their community agrees.

Data ownership, privacy and security. Seed information includes local names, farmer knowledge, locations and provenance. The Protocol states that seed in CSBs belongs to farmers and communities. Sharing controls are therefore an architectural requirement (7.7). We still need to ask participants which information should never leave the seed bank, and to agree retention and deletion rules with them.

Knowledge Hub risks. Sharing knowledge widely can help other CSBs, but public knowledge can be copied, misused or commercialised without the community's benefit, and cannot be recalled. The hub therefore defaults to private, requires deliberate consent for public entries, and separates public knowledge from seed record data. We need to ask communities which kinds of knowledge they are willing to make public, which should stay within the institution, and which should never be recorded on the platform at all. The hub also adds to the maintenance burden on volunteers, so it should be tested for real use before it grows.

Alternatives considered. A ready-made genebank system such as GGCE is powerful but built for institutions and unlikely to suit a volunteer-run CSB. A spreadsheet is easy but, as the Crop Trust expert warned, risks the loss and unintended changes typical of Excel workflows. Paper cards are familiar and need no device. The pilot compares all three.

Legal context. The 2025 High Court ruling improved the position of community seed sharing but is under appeal, and a participant's belief about legal requirements differs from the literature. Both need verification before we advise CSBs on legal matters.

9. RECOMMENDATIONS

1. Ask GeRRI what a specialist needs to see in a support request to decide on it, so that the limited view (only records linked to a request) is enough to act on.
2. Run a comparative pilot at two or three sites with different levels of institutional support, testing three formats: a paper card, a spreadsheet and the Mbegu Link app and PWA. Measure completion time, error rate, missing fields, retrieval time, continued use over time, training required, device availability, data loss and user preference. Also measure record completeness, repeated testing and time to raise a concern, and check whether the record freshness fields (7.2) make stale data easier to recognise.
3. Run the test cases in 7.14 before the pilot starts.
4. Agree community-defined values with custodians and coordinators for the germination threshold and the maximum seed-holding period, rather than importing FAO genebank thresholds.
5. Define a support pathway with GeRRI and the seed networks that names who a CSB contacts and what information to send.
6. Set data ownership, consent, sharing, retention and deletion rules with communities before any data is shared, and ask which information must never be shared. Do the same for the Knowledge Hub: ask which knowledge communities would make public, keep within their institution, or never record, and agree attribution and withdrawal rules.
7. Verify claims about legal requirements and about how GeRRI's public records handle community identifiers.

10. CONCLUSION

Our research shows a seed system in which the community layer does real conservation work with simple, resourceful methods, but leaves little record of what it holds or how it is doing. Identity lives on bottle labels, testing is repeated at every transfer, backup is informal and unrecorded, and the route to the national genebank is unclear to most seed custodians. The formal side has strong systems for identity, storage and viability, though it shares only part of this publicly, and its experts point to identity preservation between institutions as the key link to examine.

We therefore propose starting small: a community-controlled seed record, a seed movement workflow and a support-request pathway that build on existing practice, keep local names first, show how fresh and how reliable each record is, and protect community ownership by design. The MVP will succeed if a custodian can securely register and update a threatened seed variety on the mobile app, if coordinators can track identity, condition, testing history, custody and movement while managing requests, and if a CSB can flag a threatened variety

and work with GeRRI specialists on safety duplication or longer-term conservation before the seed is lost. The seed's story should travel with the seed.

The hardest problems are not the software itself but identity, data consistency, sustained data entry and institutional acceptance. Whether an app is better than paper or a spreadsheet, and whether anything beyond manual recording is warranted, are questions for the pilot and the remaining field evidence, not assumptions. Because the findings come from a limited set of sources, they should be treated as evidence from the participants and sites included, and will be refined as the research continues.

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GLOSSARY

Term	Definition
Community Seed Bank (CSB)	A place where community seeds are stored for future use and conservation.

Participatory Ecological Land Use Management Kenya (PELUM)	A prominent national network of Civil Society Organizations (CSOs) and NGOs that promotes agroecology, sustainable land use, and ecological practices among small-scale farmers and pastoralists across all 47 counties in Kenya.
GRIN-Global Community Edition (GGCE)	The historic Australian non-profit focused on preserving traditional non-hybrid food varieties inside home gardens and community networks.GRIN-Global Community Edition. It is a freely accessible, open-source software package explicitly designed for genebanks to manage, track, and document plant genetic resources in their ex-situ collections
Seed Savers Network	A non governmental organization dedicated to preserving indigenous seeds, protecting agrobiodiversity, and promoting sustainable agroecological practices among smallholder farmers in Kenya.
GRIN-Global	A powerful, highly customizable, relational database software suite designed for plant genebanks.
Genesys	The Genesys PGR Portal is a massive, centralized global repository website that has information on different seed accensions and their records.
Crop Trust	Works to protect local crop diversity, support the national seed bank, and improve food security in changing climate.

Community Sustainable Agriculture and Healthy Environmental Program (CSHEP)	A registered community-based organizatIon (CBO) and grassroots initiative
Genetic Resources Research Institute (GeRRi)	Operating under the Kenya Agricultural and Livestock Research Organization (KALRO), it serves as Kenya's **National Genebank**.